Practice workbook · Angle between two sloping faces with the cosine rule

Finding the angle between two slanted faces of a solid by dropping perpendiculars from each face onto their common edge and using the cosine rule.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-17

1. Question 8 · part e · 3 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part e only.

8

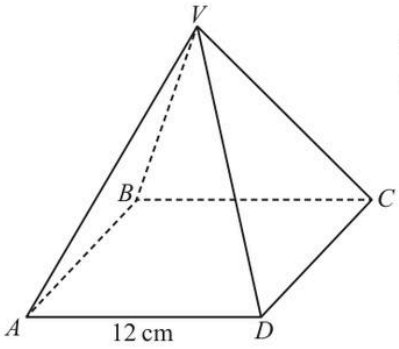


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right pyramid with vertex V and square base, $ABCD$, of side 12 cm

The size of the angle between the plane VAB and the base $ABCD$ is $30°$

(a) Show that the height of the pyramid is $2\sqrt{3}$ cm

(2)

(b) Find, in cm, the exact length of VA

(3)

(c) Find, in cm^2, the exact area of triangle VAD

(3)

(d) Find, in cm, the exact length of the perpendicular from D to VA

(2)

(e) Find, in degrees to one decimal place, the size of the obtuse angle between the plane VAB and the plane VAD

(3)

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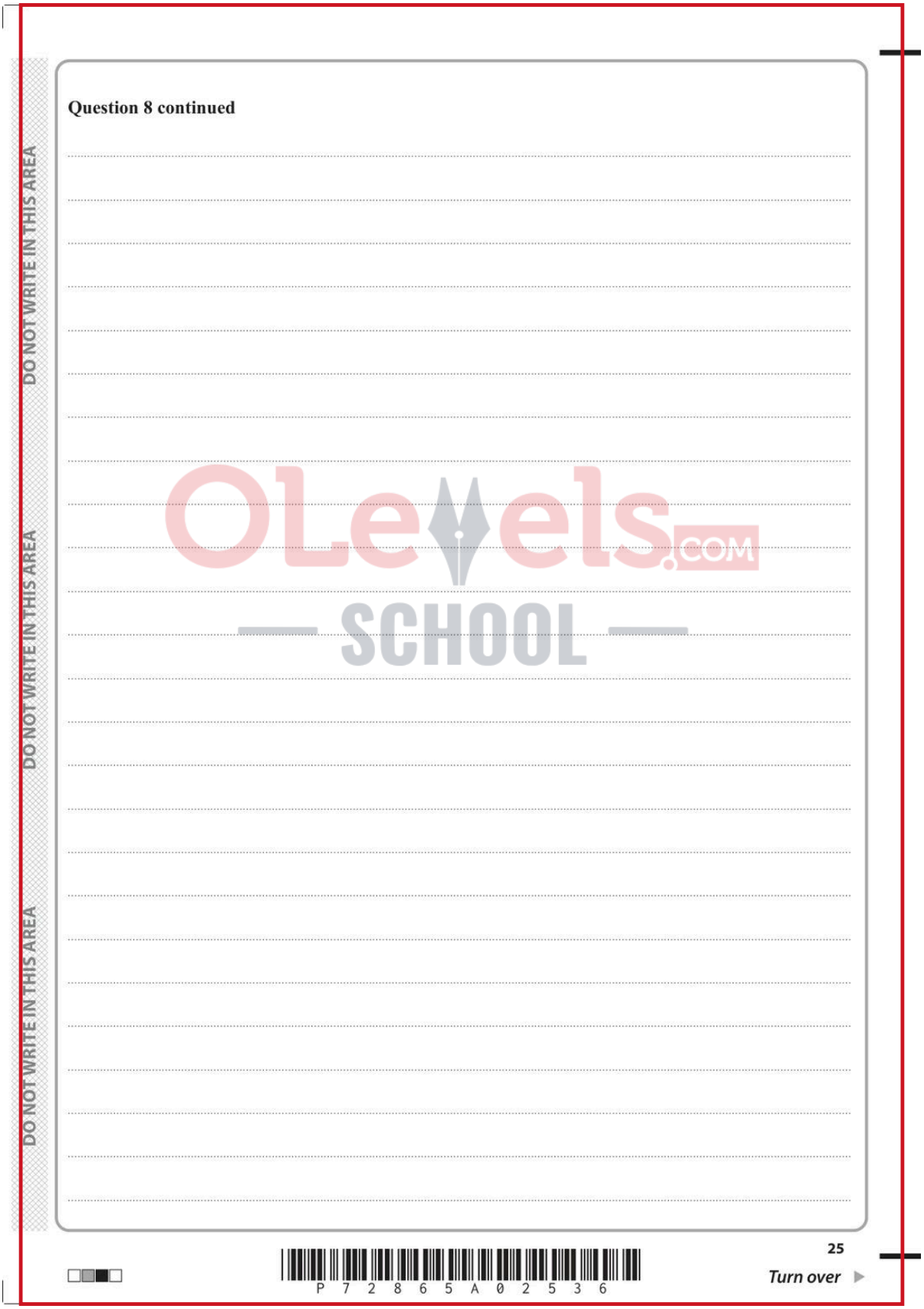
24

P 7 2 8 6 5 A 0 2 4 3 6

1. Question 8 · part e · 3 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part e only.



1. Question 8 · part e · 3 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part e only.

Question 8 continued

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1. Question 8 · part e · 3 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part e only.

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Question 8 continued

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(Total for Question 8 is 13 marks)

1. Question 8 · part e · 3 marks · Working and answer

Working and answer

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2. Question 8 · part d · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part d only.

8

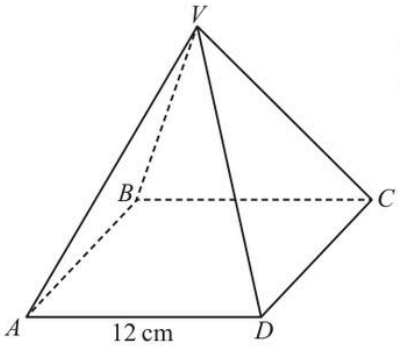


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2. Question 8 · part d · 2 marks · Working and answer

Working and answer

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3. Question 11 · part c · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

11

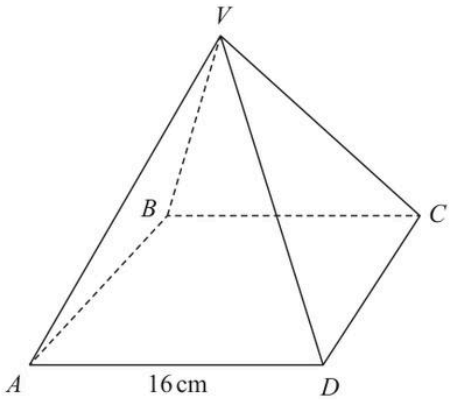


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a right pyramid with vertex V and square base, $ABCD$, of side 16 cm.

The size of angle AVC is 90°

(a) Show that the height of the pyramid is $8\sqrt{2}$ cm.

(4)

(b) Find, in cm, the length of VA.

(3)

(c) Find, in cm, the exact length of the perpendicular from D onto VA.

(3)

Find, in degrees to one decimal place, the size of

(d) the angle between the plane VAB and the base $ABCD$,

(3)

(e) the obtuse angle between the plane VAB and the plane VAD.

(3)

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3. Question 11 · part c · 3 marks · Working and answer

Working and answer

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4. Question 11 · part e · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part e only.

11

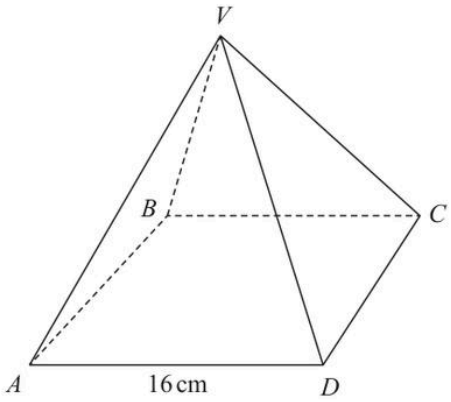


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(3)

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(3)

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P 5 8 3 7 3 A 0 3 0 3 2

4. Question 11 · part e · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part e only.

Question 11 continued

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4. Question 11 · part e · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part e only.

Question 11 continued

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(Total for Question 11 is 16 marks)

4. Question 11 · part e · 3 marks · Working and answer

Working and answer

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5. Question 4 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

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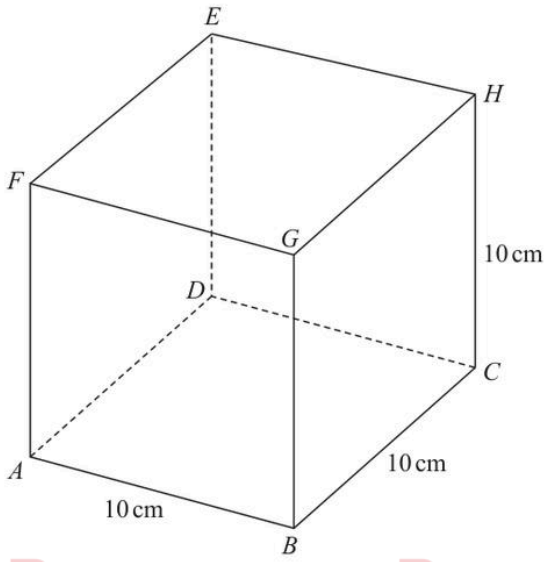


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a solid cube $ABCDEFGH$ with sides of length 10 cm.

(a) Find, in cm, the exact length of FC

(2)

(b) Find, in degrees to one decimal place, the size of $\angle FCA$

(2)

(c) Find, to the nearest degree, the size of the obtuse angle between the plane BFH and the plane BHC

(5)

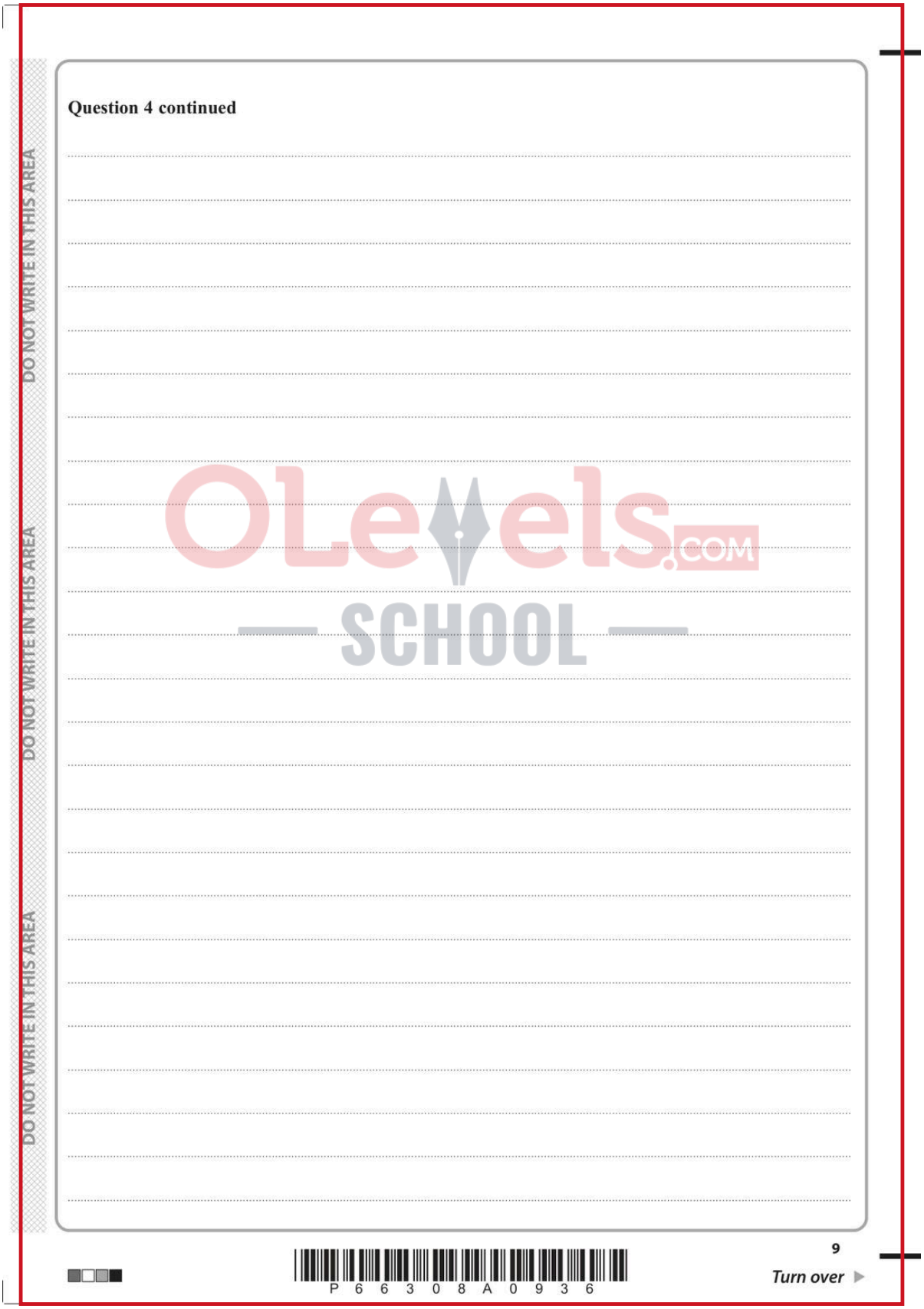
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5. Question 4 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.



5. Question 4 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

Question 4 continued

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5. Question 4 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 9 marks)

5. Question 4 · part c · 5 marks · Working and answer

Working and answer

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6. Question 10 · part d · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

10

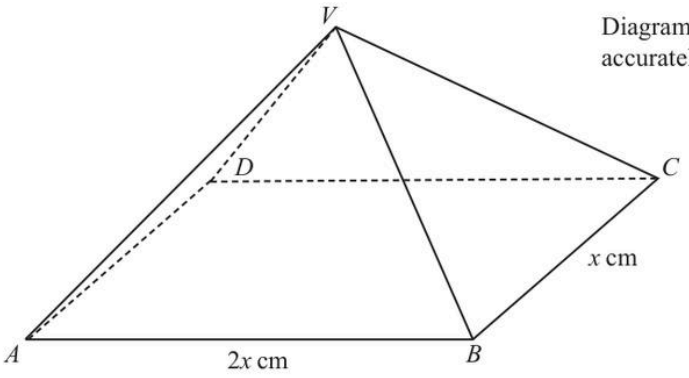


Figure 4

Figure 4 shows a right pyramid $ABCDV$

The base of the pyramid is a rectangle where,

$AB = DC = 2x$ cm $AD = BC = x$ cm

The edges VA, VB, VC and VD are all of equal length.

The angle between VA and $ABCD$ is 45°

(a) Show that $VA = \frac{\sqrt{10}}{2}x$ cm

(3)

(b) Find in cm, the exact height of the pyramid in terms of x

(2)

Find, in degrees to one decimal place,

(c) the size of angle VBA

(2)

(d) the size of the obtuse angle between the plane AVC and the plane BVD

(4)

Given that the volume of the pyramid is $9\sqrt{5}$ cm^3

(e) find the value of x

(2)

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6. Question 10 · part d · 4 marks · Working and answer

Working and answer

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7. Question 11 · part c · 6 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

11

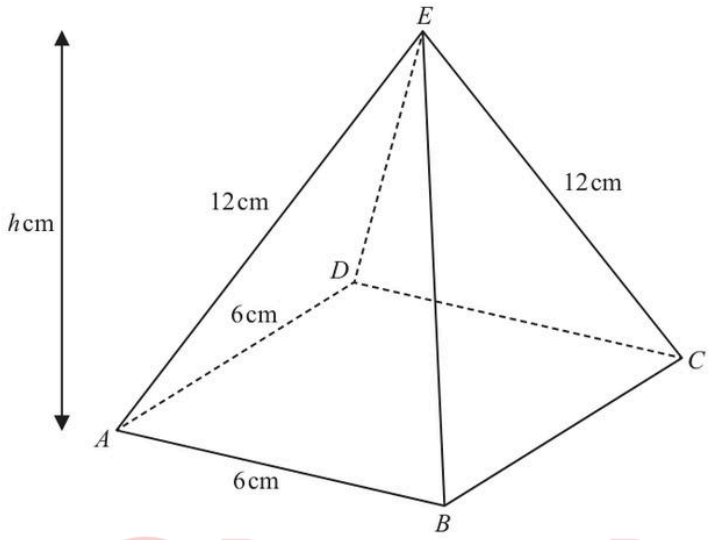


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a right pyramid $ABCDE$ with

- vertex E
- horizontal square base $ABCD$ of side 6 cm
- $EA = EB = EC = ED = 12$ cm
- vertical height of h cm

(a) Show that $h = 3\sqrt{14}$

(3)

(b) Find, in degrees to one decimal place, the size of the angle between EB and the plane $ABCD$

(2)

The midpoint of AE is P and the midpoint of BE is Q

(c) Find, to the nearest degree, the size of the obtuse angle between the plane EPQ and the plane $PQCD$

(6)

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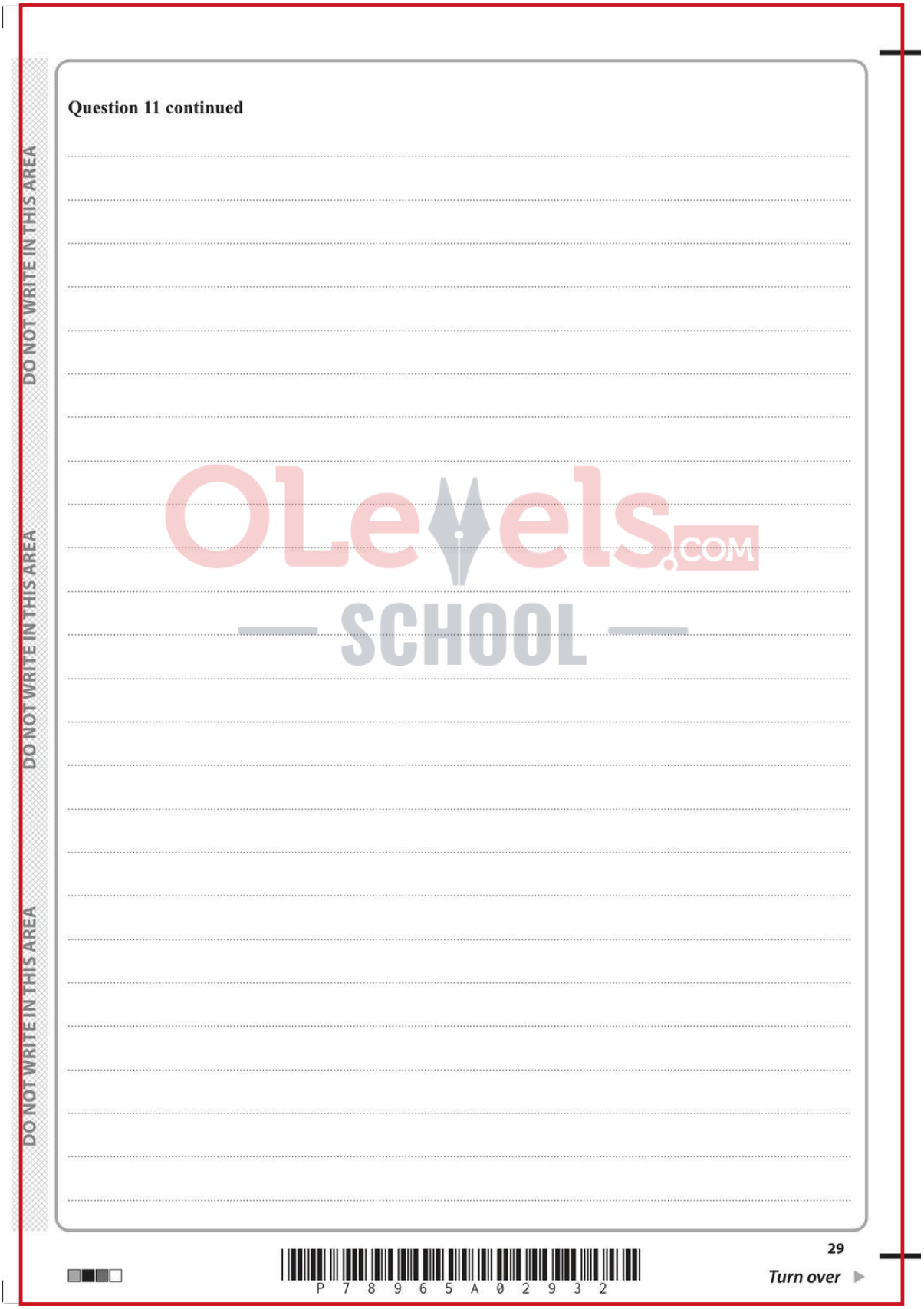
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7. Question 11 · part c · 6 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.



7. Question 11 · part c · 6 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

Question 11 continued

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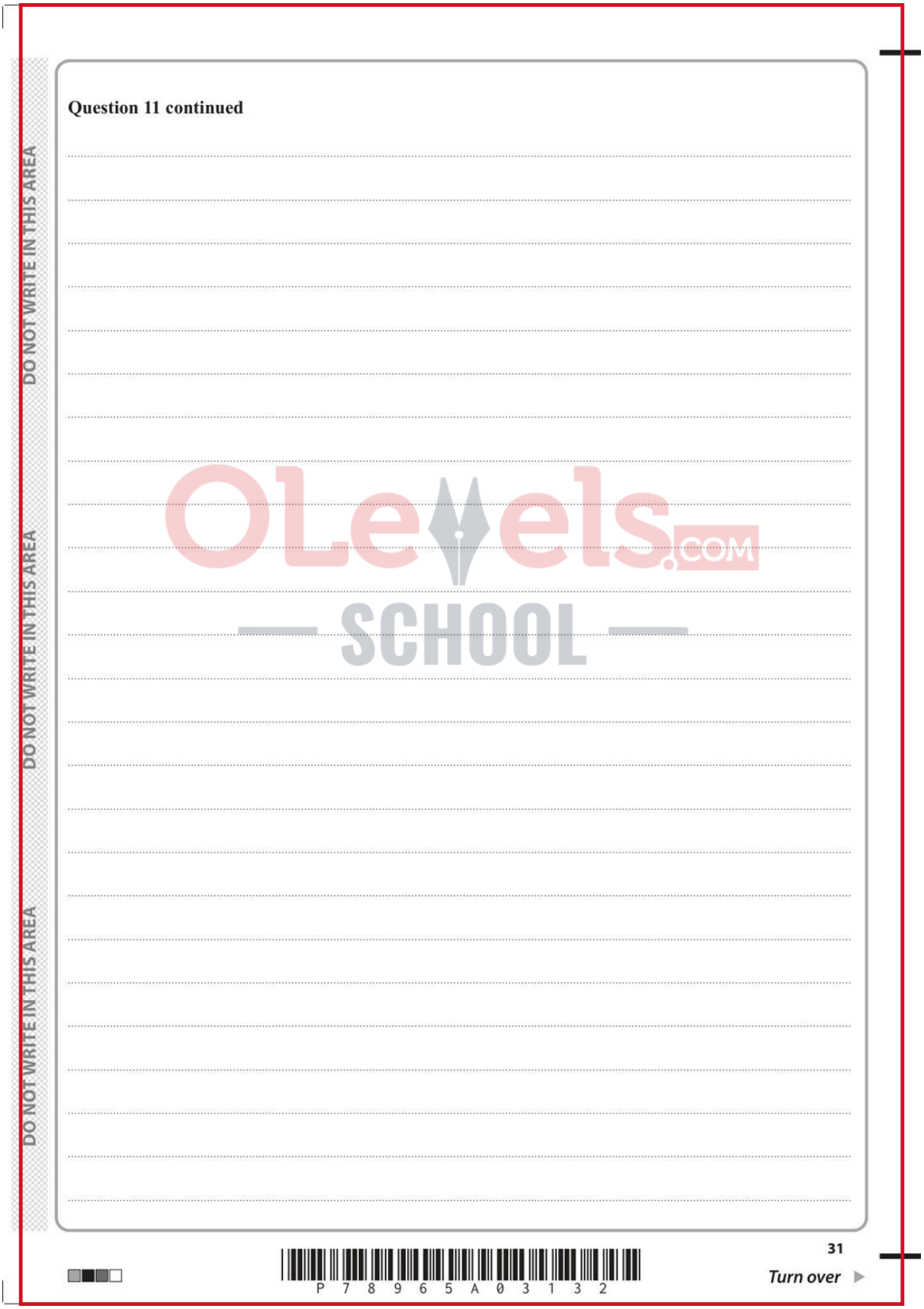
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7. Question 11 · part c · 6 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.



7. Question 11 · part c · 6 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

Question 11 continued

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(Total for Question 11 is 11 marks)

7. Question 11 · part c · 6 marks · Working and answer

Working and answer

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8. Question 9 · part c · 4 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

9

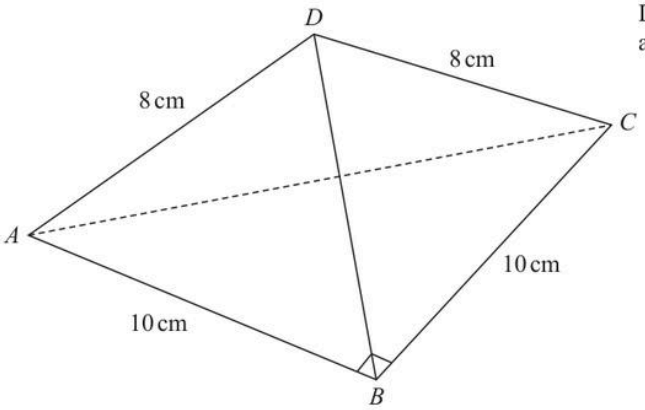


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a triangular pyramid $ABCD$ with base ABC

$AB = BC = 10$ cm $\quad AD = CD = 8$ cm $\quad \angle ABC = 90°$

(a) Find the exact length of AC

Give your answer in the form $p\sqrt{q}$ cm where p is an integer and q is a prime number.

(2)

The point M is the midpoint of AC

(b) Find the exact length of BM

Give your answer in the form $m\sqrt{n}$ cm where both m and n are prime numbers.

(2)

Given that $BD = 6$ cm,

(c) find, in degrees to one decimal place, the size of the acute angle between the plane ACD and the plane ABC

(4)

The base ABC of the pyramid is placed on a horizontal plane.

(d) Find, in cm to 3 significant figures, the vertical height of D above the base.

(2)

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8. Question 9 · part c · 4 marks · Working and answer

Working and answer



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9. Question 9 · part d · 4 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

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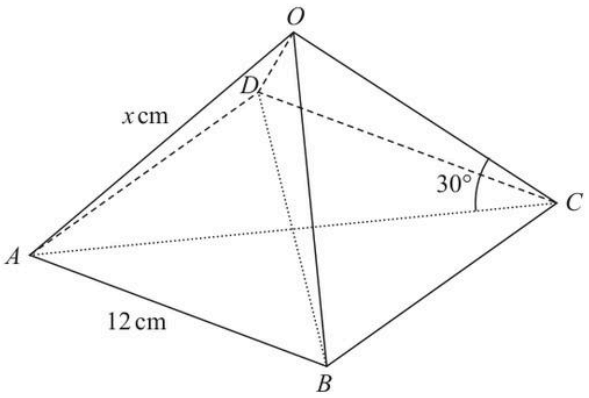


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows the right pyramid $OABCD$ with a square base $ABCD$ of side 12 cm.

$OA = OB = OC = OD = x$ cm and $\angle OAC = \angle ODB = \angle OCA = \angle OBD = 30°$

(a) Find the exact length of AC

(2)

(b) Show that $x = 4\sqrt{6}$

(2)

(c) Find the total surface area, to the nearest cm^2, of the pyramid.

(5)

(d) Find the size of the obtuse angle, to the nearest degree, between the plane OAB and the plane OBC

(4)

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9. Question 9 · part d · 4 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 9 continued

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9. Question 9 · part d · 4 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 9 continued

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9. Question 9 · part d · 4 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 9 continued

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(Total for Question 9 is 13 marks)

9. Question 9 · part d · 4 marks · Working and answer

Working and answer

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